

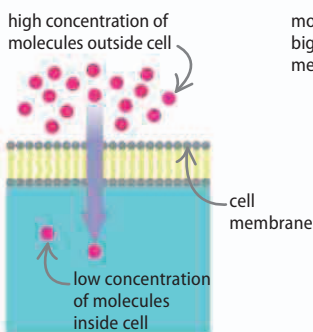
Cells at work

EACH CELL IS LIKE A MICROSCOPIC FACTORY.

All the processes needed for life, such as releasing energy from food, removing waste materials, and growth, take place inside cells.

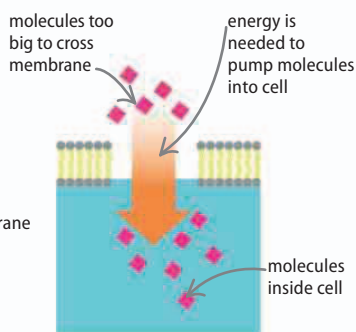
Cell transport

Cells process a wide range of chemicals. Inside the cell, large molecules such as proteins and even entire organelles are hoisted around by microtubules, which are also used in cell division. Some chemicals must be moved between organelles inside the cell, and others travel in and out through the cell membrane. Here are the main ways substances enter cells.



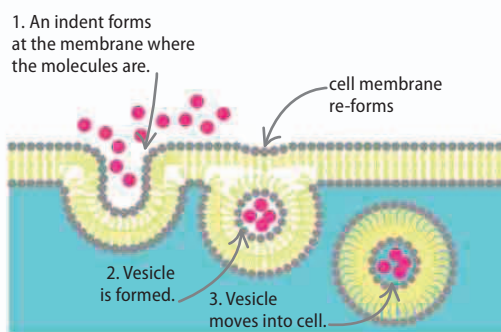
△ Diffusion

Diffusion happens when a substance spreads out, moving from areas of high concentrations to low.



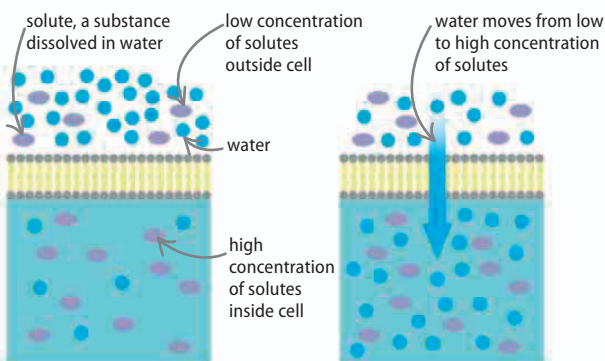
△ Active transport

If a molecule is too big or is unable to dissolve in the cell membrane, it is moved across in a process that uses energy.



△ Endocytosis

If molecules are too big to be pumped into a cell by active transport, a cell uses energy to put them in a sac, called a vesicle. This vesicle is formed from the cell membrane, and breaks open to release its contents once inside. When a cell moves a vesicle of material out, it is called exocytosis.



△ Osmosis

Osmosis is a type of liquid diffusion that takes place when solutions are separated by a membrane. Large dissolved molecules are blocked from diffusing into the cell. Instead, the water balances both sides, by moving from the low concentration side to the high.

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Bacteria cells can divide every 20 minutes, and one germ can grow to four billion trillion in 24 hours.

REAL WORLD

Wilted flowers

Osmosis creates a force that moves water in and out of cells. When cut flowers are placed in freshwater, water floods into the plant cells by osmosis, making them full and rigid. When the water has gone, osmosis pulls the water out of the cells. The water evaporates, and the flowers wilt.

